Table of Contents

CellCraze — End-to-End Business Workflow Simulation & Feature-Gap Audit

Date: 2026-10-23 · **Auditor:** Claude Code (QA/BA/WooCommerce/POS/inventory) **Repo:** `harshawalisundara97/CellCraze-Mobile-wordpress` · **Branch:** `main` · **HEAD:** `d58cea5` **This report is the main source of truth for the audit.** No application code, DB, config, or Git history was changed in producing it (confirmed in §13).

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1. Environment & test limitations (read first)

Item	Finding
Session	**Cloud clone** (Linux container), not the Windows/XAMPP laptop.
Working tree	`main` @ `d58cea5`, **clean**.
Running website	**None reachable.** No web server in this env; the live site `phoneshop.nuvirahub.com` cannot be reached from here.
Browser automation	**Unavailable** — no Chromium/Chrome installed, and no site to drive. **All UI simulations are NOT RUN.**
Database	**Unavailable** — no MySQL, port 3306 not listening, no DB container. **All DB-state checks are BLOCKED.** (The laptop's MariaDB Aria/`mysql.plugin` issue can only be diagnosed on the laptop.)
Laptop-only inputs	The laptop's **uncommitted GRN diff** in `class-cellcraze-purchasing.php` and docs `CELlCRAZE-EXISTING-FEATURES-AUDIT.md`, `CELLCRAZE-FEATURE-GAP-IMPLEMENTATION-PLAN.md`, `F16-IMPLEMENTATION-KICKOFF.md` are **not present in this clone** and were not inspected.

Method actually used: rigorous **source-code audit** of the committed `cellcraze-core` plugin and `cellcraze` theme, cross-referenced with `docs/FEATURE-TRACEABILITY.md`, `docs/CELLCRAZE-MASTER-IMPLEMENTATION-ROADMAP.md`, and the task specs. Findings are labelled **CONFIRMED** (proven in source), **SUSPECTED** (needs runtime), or **BLOCKED** (needs DB/UI). No workflow is marked "passed" — nothing was executed at runtime.

No credentials, secrets, or customer data were accessed or exposed.

2. Workflow-by-workflow results (source-based)

- **Catalogue/PDP/cart/checkout/accounts** — present in theme + Woo core (CUS-01..13). Variants not built (products `simple`); storefront brand/price/storage **filters not wired** (sort only).
- **Payments / bank-slip** — COD + BACS (Woo) + custom **bank-slip upload/verify** (`class-cellcraze-bankslip.php`): nonce, ownership, size, extension + real-MIME, private gated store, **slip ≠ paid**, idempotent verify. **Strong.** No online card gateway.
- **Serialized IMEI** — `wp_cellcraze_imei` with states available/reserved/sold/damaged/returned; **atomic compare-and-set claim**; idempotent, concurrency-safe FIFO allocation; audit-logged. **Strong.**
- **Accessory qty** — WooCommerce stock (single writer). OK.
- **Stock adjustments** — **no reasoned/audited manual-adjustment workflow**; IMEI delete path has an **unaudited** decrement (see SIM-02).
- **Suppliers / PO / GRN** — `_supplier`, `_po/_po_item/_grn/_grn_item`; partial receiving reuses `IMEI_Admin::receive()` + `wc_update_product_stock`; `grn_token` idempotency; over-receive guard. **Strong** (⚠️ reconcile laptop WIP).
- **Expenses / COGS / dashboard** — expense CPT-like table + audit; COGS captured on processing/ completed (serialized read live from IMEI cost, accessory snapshot + missing-cost flag); dashboard shows real KPIs. **COGS not reversed on refund** (SIM-06).
- **POS** — **not implemented** (only REST `/imei/available`, `/imei/assign`, `/channel` + channel tagging). SIM-04.
- **Roles/caps** — server-side caps (`cellcraze_use_pos/receive_stock/manage_imei/manage_expenses/view_reports`), roles cashier/warehouse/order-processing. **Strong.** Missing `cellcraze_adjust_stock` (F16).
- **REST / tables** — 3 routes, permission-gated; 8 custom tables (dbDelta additive).
- **Delivery/courier/COD** — **not implemented.** SIM-07.
- **Returns/refunds/RMA** — **not implemented**; worse, **refund auto-restores IMEI to available** (SIM-01).
- **Notifications/cron** — Woo transactional email only; no SMS/WhatsApp; no low-stock alerting.
- **Responsiveness** — theme has responsive CSS (not runtime-verified).

3. Test matrix (Scenarios A–J)

Legend: **NOT RUN** (needs browser), **BLOCKED** (needs DB), **SRC-VERIFIED** (behavior proven in source), **SRC-DEFECT** (defect proven in source).

#	Scenario	Status	Note
A	Customer buys accessory online	NOT RUN	Flow exists in theme+Woo; qty

#	Scenario	Status	Note
			validation = Woo core.
B	Customer buys phone online (IMEI alloc)	SRC-VERIFIED (no-double-sell) / NOT RUN (UI)	Atomic claim + idempotent allocation proven (CC-03/04).
C	Bank transfer + slip verification	SRC-VERIFIED / NOT RUN (UI)	Slip≠paid, idempotent verify, gated (CC-09).
D	Owner manages products/inventory	NOT RUN / SRC-DEFECT	IMEI delete unaudited decrement (SIM-02); no reasoned adjust (SIM-03).
E	Purchasing → PO → GRN	SRC-VERIFIED / NOT RUN (UI)	Partial/ idempotent/over-receive guard (ADM-10); reconcile laptop WIP.
F	Physical POS sale	NOT RUN — **MISSING**	No POS register/sale/receipt /reconciliation (SIM-04).
G	Returns / refunds / RMA	NOT RUN — **MISSING + SRC-DEFECT**	No RMA; refund auto-restocks IMEI (SIM-01); COGS not reversed (SIM-06).
H	Delivery / COD	NOT RUN — **MISSING**	No courier/tracking/COD remittance (SIM-07).
I	Reporting / reconciliation	BLOCKED / NOT RUN	Dashboard real KPIs; valuation/reconciliation report missing (SIM-12).
J	Security & usability	SRC-VERIFIED (partial)	Caps/nonces/REST gating/escaping/prepared SQL present; UI a11y NOT RUN.

4–8. Findings (full detail)

Each finding also appears in the consolidated register (§ Issue Register). Categories: BUG / MISSING / INCOMPLETE / ENHANCEMENT / SECURITY / DATA INTEGRITY / UX / DEPENDENCY.

SIM-01 — Refund automatically returns an IMEI to sellable stock

- **Category/Severity:** DATA INTEGRITY · **P1** (P0 if refunds are used before RMA exists). **Confirmed.**
- **Repro (source):** `class-cellcraze-imei-orders.php` hooks `woocommerce_order_status_refunded` and `woocommerce_order_fully_refunded` → `restore_order()` → `IMEI_DB::restore()` sets status = `available`.
- **Expected:** a refunded phone must **not** become sellable automatically; it should be quarantined (`returned`) pending an explicit, audited inspection/regrade.
- **Actual:** refund silently makes the unit `available` again (audited, but wrong disposition).
- **Impact:** a device that may be damaged/not-returned can be resold; stock integrity + warranty risk.
- **Files:** `class-cellcraze-imei-orders.php` (`restore_order`, hook registration lines ~26–28).
- **Fix:** on refund, transition to `returned` (not `available`); only RMA inspection (`re_grade`, manager) returns it to `available`. Keep cancel→restore (unsold) as-is or review.
- **Dependencies:** RMA-02 (regrade) · decision D5. **Risk:** changing refund behavior affects existing flows — needs tests.
- **Acceptance:** [ ] refund sets IMEI `returned`, not `available`; [ ] only audited regrade makes it sellable; [ ] cancel-before-ship still restores. **Regression:** CC-04 lifecycle stays green.
- **Task:** feeds **RMA-02** (and interim guard could land with **F16-02**).

SIM-02 — Unaudited stock decrement when deleting an available IMEI

- **Category/Severity:** DATA INTEGRITY · **P0**. **Confirmed.**
- **Repro (source):** `class-cellcraze-imei-admin.php::handle_delete_imei()` → for an `available` unit calls `self::adjust_stock($product_id, -1)` with **no `Cellcraze_Audit::log()`** entry.
- **Expected:** every stock movement is audited (actor, reason, signed delta, reference).
- **Actual:** sellable stock drops by one with no audit trail; gated only by `manage_woocommerce`.
- **Impact:** silent stock loss; breaks reconciliation; no accountability.
- **Files:** `class-cellcraze-imei-admin.php` (`handle_delete_imei`, `adjust_stock`).
- **Fix:** route through the F16 audited-adjustment service with a reason + `cellcraze_adjust_stock` cap. **Acceptance:** [ ] delete writes exactly one audit row +

one decrement; [ ] cap enforced; [ ] sold units still undeletable. **Task: F16-03** (needs F16-01/02).

SIM-03 — No capability-gated, reasoned, bounded manual stock adjustment

- **Category/Severity:** MISSING (DATA INTEGRITY) · **P1**. **Confirmed.**
- **Evidence:** grep shows only a private `adjust_stock()`; no `cellcraze_adjust_stock` cap; no reasons (`miscount/damaged/lost/display_unit/write_off/re_grade/found`), no magnitude cap, no negative-stock guard, no manager-only restriction for loss/write-off.
- **Impact:** inventory corrections are either impossible or unsafe/unaudited.
- **Fix:** F16-01 (cap+reasons+validation) + F16-02 (audited service + IMEI-state guards).
- **Acceptance:** see F16 specs. **Tasks: F16-01, F16-02**.

SIM-04 — POS (physical shop) not implemented

- **Category/Severity:** MISSING · **P1**. **Confirmed.**
- **Evidence:** only REST `/imei/available`,`/imei/assign`,`/channel` + `cellcraze_use_pos` cap + channel tagging exist; no register UI, sale flow, receipt, or cash reconciliation.
- **Impact:** no in-store selling through the shared inventory.
- **Fix:** POS-01 (sale lifecycle = Woo order, channel=pos, IMEI alloc, audit) → POS-02 (receipt) → POS-03 (register/reconciliation/cashier). **Dependency:** D2 (custom vs plugin), F16-02, PAY-01.
- **Tasks: POS-01..04**.

SIM-05 — Returns / refunds / RMA workflow missing

- **Category/Severity:** MISSING · **P1**. **Confirmed.**
- **Evidence:** `returned` status exists but no intake/inspection/refund workflow; refund path mishandles disposition (SIM-01).
- **Fix:** RMA-01 (intake + limits + IMEI validation) → RMA-02 (inspection/regrade, audited) → RMA-03 (refund + COGS/profit reversal, idempotent). **Dependency:** D5, D6.
- **Tasks: RMA-01..03**.

SIM-06 — COGS/profit not reversed on refund

- **Category/Severity:** DATA INTEGRITY · **P1**. **Confirmed.**
- **Evidence:** `class-cellcraze-cogs.php` captures on processing/completed only; no reversal hook on refund. Profit after refunds will be overstated.
- **Fix:** reversal coordinated in RMA-03 + reconciliation in REPORT-01; define method (D6).
- **Tasks: RMA-03, REPORT-01**.

SIM-07 — Courier / delivery / COD remittance missing

- **Category/Severity:** MISSING · **P2** (P1 for a delivery business). **Confirmed.**
- **Evidence:** no shipment model, tracking, or COD collection/remittance.
- **Fix:** SHIP-01 (manual courier + tracking + statuses) → SHIP-02 (COD reconciliation). **Dependency:** D4. **Tasks: SHIP-01, SHIP-02**.

SIM-08 — Online card gateway not available

- **Category/Severity:** DEPENDENCY · **P1**. **Confirmed.**
- **Evidence:** no gateway plugin; by design, card is live-only. Blocked on **D1**.
- **Fix:** PAY-GW-01 after provider pick; sandbox then live; dup-webhook guard. **Task: PAY-GW-01**.

SIM-09 — Storefront variants + filters incomplete

- **Category/Severity:** INCOMPLETE · **P2**. **Confirmed.**
- **Evidence:** attributes Brand/Storage/Colour exist; products are `simple` (no variations); shop filters not wired (sort only) — `docs/FEATURE-TRACEABILITY.md` CUS-05/06.
- **Fix:** variable phones + brand taxonomy + filter blocks. **Task: T-105** (existing ID).

SIM-10 — Low-stock / reorder alerts missing

- **Category/Severity:** MISSING · **P2**. **Confirmed.**
- **Evidence:** dashboard shows a low-stock count; no alerting/digest. **Task: INV-03**.

SIM-11 — Content pages (contact/FAQ) missing; policy pages unbranded

- **Category/Severity:** UX/INCOMPLETE · **P2/P3**. **Confirmed.** (CUS-14/15.)
- **Fix:** add Contact/FAQ/About; brand+link policy pages. **Task:** new content task (P3) or fold into storefront polish.

SIM-12 — Inventory valuation / reconciliation report missing

- **Category/Severity:** ENHANCEMENT (DATA INTEGRITY support) · **P1**. **Confirmed.**
- **Evidence:** cost captured; movement ledger built (T-101); **valuation + reconciliation report not built** (ADM-11 partial). **Task: REPORT-01** (D6 for method).

Preserve (implemented correctly — do NOT rebuild)

CC-03 atomic IMEI claim · CC-04 IMEI lifecycle · CC-06 receiving↔stock sync · CC-09 bank-slip (slip≠paid, idempotent, gated) · CC-13 roles/caps · ADM-05 channel tagging · ADM-08 dashboard · ADM-09 suppliers · ADM-10 PO/GRN (idempotent, over-receive guard) · T-101 stock-movement/audit ledger · storefront CUS-01..13.

Security notes (Scenario J, source)

Bank-slip + expense uploads: nonce, ownership, size, extension + real-MIME, private gated store — **good**. REST routes have real permission callbacks. Admin actions use `check_admin_referer` and caps. SQL uses `$wpdb->prepare`. Output escaped. **Residual:** IMEI delete cap is coarse (`manage_woocommerce`) and unaudited (SIM-02); harden via F16. No destructive security tests were run.

Issue Register (consolidated)

ID	Cat	Title	Sev	Module	State	Expected	Actual	Impact	Files/Fn	Fix	Deps	Acceptance	Task
SIM	DAT	Ref	P1	Ret	Conf	Qua	Set	Rese	imei	Set	RM	refu	RM

ID	Cat	Title	Sev	Module	State	Expected	Actual	Impact	Files/Fn	Fix	Deps	Acceptance	Task
-01	A INTEGRITY	und auto-restocks IMEI as available		urns/IMEI	irmed	rantineas returned, regrade to sell	available on refund	ll bad unit	-orders restore_order	returned; regrade via RMA	A-02, D5	nd→returned; regrade-only sellable	A-02
SIM-02	DATA INTEGRITY	Unaudited decrement on available-IMEI delete	P0	Inventory/IMEI	Confirmed	Audited, cap-gated decrement	Unaudited −1	Silent stock loss	imei-admin handle_delete_imei	Route via F16 audited service	F16-01/02	1 audit row + 1 decrement; cap	F16-03
SIM-03	MISSING	No reasoned/bounded audited stock adjustment	P1	Inventory	Confirmed	Cap+reasons+limits+audit	None	Unsafe/absent corrections	roles, new service	Build F16 adjust	INV-01	per F16 specs	F16-01/02
SIM-04	MISSING	POS not implemented	P1	POS	Confirmed	Full POS on shared	REST+tag only	No in-store sale	new POS controller	Build POS	D2,F16-02,PAY-01	POS == Woo order+I	POS-01..04

ID	Cat	Title	Sev	Module	State	Expected	Actual	Impacts	Files/Fn	Fix	Deps	Acceptance	Task
						inventory						MEI+audit	
SIM-05	MISSING	Returns/RMA workflow absent	P1	Returns	Confirmed	Intake/inspect/refund	None	No returns handling	new RMA	Build RMA	D5, D6	per RMA specs	RMA-01..03
SIM-06	DATA INTEGRITY	COGS/profit not reversed on refund	P1	Accounting	Confirmed	Refund reverses COGS	No reversal	Profit overstated	cogs capture	Reverse in RMA-03	D6	profit reconciles post-refund	RMA-03/REPORT-01
SIM-07	MISSING	Courier/COD remittance absent	P2	Delivery	Confirmed	Dispatch+track+COD recon	None	No delivery ops	new SHIP	Build SHIP	D4	per SHIP specs	SHIP-01/02
SIM-08	DEPENDENCY	Online card gateway absent	P1	Payments	Confirmed	Gateway sandbox+live	None	No online card	Woo gateway	After provider	D1	sandbox paths pass	PAY-GW-01
SIM-09	INCOMPLE	Variants+	P2	Storefront	Confirmed	Variable pho	simple+sort	Weaker UX	theme/attr	Build T-105	—	variant pric	T-105

ID	Cat	Title	Sev	Module	State	Expected	Actual	Impact	Files/Fns	Fix	Deps	Acceptance	Task
	TE	shop filters				nes+filters	only		s			e/stock; filters narrow	
SIM-10	MISSING	Low-stock/reorder alerts	P2	Inventory	Confirmed	Alerts/digest	count only	Stockouts	dashboard/cron	Build INV-03	INV-02	low-stock flagged; email	INV-03
SIM-11	UX	Contact/FAQ missing; policy unbranded	P3	Storefront	Confirmed	Pages present/linked	missing	Trust/UX	pages	Add pages	—	pages exist+linked	(new P3)
SIM-12	ENHANCEMENT	Valuation/reconciliation report	P1	Reporting	Confirmed	Reconcile+value	missing	Blind spots	reports	Build REPORT-01	D6	reconciles to fixture	REPORT-01

Recommended implementation order (reuses existing task IDs — no competing roadmap)

1. Critical data-integrity first: PLAN-01 (finish on laptop) → INV-01 → **F16-01** → **F16-02** → **F16-03** (fixes SIM-02/03; interim guard for SIM-01) → GRN-01 (preserve laptop WIP). **2. Required missing features:** PAY-01 → **RMA-01** → **RMA-02** → **RMA-03** (fixes SIM-01/05/06) → **POS-01** → **POS-02** → **POS-03** (SIM-04). **3. Finish incomplete:** T-105 (SIM-09) → INV-02 → REPORT-01 (SIM-06/12) → INV-03 (SIM-10). **4. Enhancements:** SHIP-01 → SHIP-02 (SIM-07);

PAY-GW-01 after D1 (SIM-08). **5. Optional:** Contact/FAQ + policy branding (SIM-11); POS-04. **Gate QA-01 (regression/security/HPOS/recovery) before release.**

Regression test plan

Keep green: CC-03/04/06/09/13, ADM-05/08/09/10, T-101. For every inventory/payment/IMEI/refund change add: duplicate-submission, partial-failure, concurrency, and reconciliation tests. Testing adapted to repo (no PHPUnit assumed): `php -l`, WP-CLI `eval-file`, documented HTTP/browser scripts; **UI tests remain NOT RUN until a browser+running site is available; DB tests BLOCKED until MySQL is healthy.**

Unresolved decisions (block specific tasks)

D1 gateway→PAY-GW-01 · D2 POS approach→POS-01..04 · D4 courier→SHIP · D5 RMA approach→RMA + SIM-01 fix · D6 COGS method→RMA-03/REPORT-01 · D7 tax · D8 warranty · D9 single-shop (confirmed) · D10 SMS/WhatsApp. (See `docs/CELLCRAZE-DECISIONS-AND-OPEN-QUESTIONS.md`.)

Proposed documentation changes (no code)

- Add **SIM-01** (refund→auto-sellable) and **SIM-06** (COGS not reversed on refund) to the risk register in the master roadmap and tie the fixes into **RMA-02/RMA-03** acceptance.
- In `CELLCRAZE-IMPLEMENTATION-STATUS.md`, note SIM-01/SIM-02 as **confirmed defects** feeding F16-03 / RMA-02 (evidence-based), without changing task statuses to Implemented/Tested.